Find the values of λ & μ such that the system

$$x+2y+3z=6$$
$$2x+3y+4z=8$$
$$2x+3y+\lambda z=\mu$$

has i) no solution $[\rho(A:B) \neq \rho(A)]$

ii) a unique solution $[\rho(A:B)=\rho(A)=3]$

and iii) infinitely many solutions. $[\rho(A:B)=\rho(A)<3]$

Soln: Let
$$x+2y+3z=6$$
$$2x+3y+4z=8$$
$$2x+3y+\lambda z=\mu$$
be the given system.

$\therefore$ we have
$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 2 & 3 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ \mu \end{bmatrix}$$
$$A \quad X = B$$

When $A=\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 2 & 3 & \lambda \end{bmatrix}$, $X=\begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B=\begin{bmatrix} 6 \\ 8 \\ \mu \end{bmatrix}$,

Consider
$$(A:B)=\left[\begin{array}{ccc:c} 1 & 2 & 3 & 6 \\ 2 & 3 & 4 & 8 \\ 2 & 3 & \lambda & \mu \end{array}\right]$$

$R_3 - R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & 3 & 4 & 8 \\ 0 & 0 & \lambda-4 & \mu-8 \end{array}\right]$$

i) For no solution

$\rho(A:B) \neq \rho(A)$

$\therefore \lambda - 4 = 0$ and $\mu - 8 \neq 0$

$\boxed{\therefore \lambda = 4 \text{ and } \mu \neq 8}$

ii) For unique solution,

$\rho(A) = \rho(A:B) = 3$

$\therefore \lambda - 4 \neq 0$ for all values of μ.

$\boxed{\therefore \lambda \neq 4 \text{ for all values of } \mu}$

iii) For infinitely many solutions:

$\rho(A:B) = \rho(A) < 3$.

We get $\lambda - 4 = 0$ and $\mu - 8 = 0$

$\boxed{\therefore \lambda = 4 \text{ and } \mu = 8}$

Solve the following system

$$x+2y+3z=0$$
$$2x+3y+4z=0$$
$$3x+4y+5z=0$$

Soln: Let $x+2y+3z=0$

$$2x+3y+4z=0$$
$$3x+4y+5z=0$$

~~So~~ the given system, we have

$$\begin{bmatrix}1&2&3\\2&3&4\\3&4&5\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$AX=0$$

where $A=\begin{bmatrix}1&2&3\\2&3&4\\3&4&5\end{bmatrix}$, $X=\begin{bmatrix}x\\y\\z\end{bmatrix}$.

Consider

$$A=\begin{bmatrix}1&2&3\\2&3&4\\3&4&5\end{bmatrix}$$

R_2-2R_1, R_3-3R_1

$$\sim\begin{bmatrix}1&2&3\\0&-1&-2\\0&-2&-4\end{bmatrix}$$

R_3-2R_2

$$\sim\begin{bmatrix}1&2&3\\0&-1&-2\\0&0&0\end{bmatrix}$$

$\rho(A)=2=r<3=n$

The system has infinitely many solutions and has exactly $3-2=1$ parameter.

The reduced system is

$$x+2y+3z=0 \quad \text{—} \quad (1)$$
$$y+2z=0 \quad \text{—} \quad (2)$$

Put $\boxed{z=t}$, from eqn (2) we get

$$y=-2z=-2t$$

$\therefore$ $\boxed{y=-2t}$

from eqn (1), we get

$$x=-2y-3z=-2(-2t)-3t$$
$$x=4t-3t=t$$

$\boxed{x=t}$

The solution is given as

$$\boxed{\begin{matrix}x=t\\y=-2t\\z=t\end{matrix}}$$